

Strengthen State Capacity to Wield AI as a Geopolitical Resource

By [Noema Research](#)

In this response, we articulate policy objectives that we argue to be of strategic relevance to the US, and indeed to any global AI power that is determined to lead in the development of this remarkably powerful technology. This resource is structured as follows:

- **New Resource Emerging.** In this section, we describe AI as one of many geopolitically relevant resources, such as steel or energy.
- **Strengthen State Capacity.** In this section, we introduce two straightforward policy objectives concerning this new resource.
- **Leverage, Credibility, Security.** In this section, we reflect on the geopolitical impacts of the suggested initiatives.
- **Barriers Versus Setbacks.** Finally, we advance an intuitive framing for weighing the upsides and downsides of specific AI policies.

We conclude with pointers to further resources related to the arguments advanced in the rest of the document.

New Resource Emerging

An effective way of conceiving of AI development is as the creation of a qualitatively novel resource based on certain raw materials and manufacturing processes. Just as steel production requires iron ores and specialized furnaces, and energy generation demands fossil fuels and combustion equipment, AI development necessitates computational power and learning algorithms. Feed in larger quantities of raw materials and more efficient manufacturing processes, and you predictably obtain more "virtual labor" that can be used to extend what humans can achieve.

Perhaps more than any other resource that has preceded it, AI can be applied in an incredibly wide range of contexts. The intellectual faculties of these virtual entities may be harnessed to augment the productivity of the human workforce, may be directed

towards strengthening the cybersecurity of critical infrastructure, and may be recruited towards furthering the development of effective healthcare interventions.

Naturally, the remarkable generality of this intellectual labor forged out of computing power and learning algorithms may also be applied in ways detrimental to national interests. These virtual capabilities may disrupt the labor market in ways that cause societal unrest, may enable foreign adversaries to identify vulnerabilities in domestic infrastructure, and may empower rogue actors to interfere with public health in increasingly sophisticated ways.

Like steel, energy, and countless other resources being manufactured on an industrial scale, AI is a highly geopolitically relevant resource due to its dual-use nature. The unusual origin of this new asset — largely, the private sector — has led governments from around the world to scramble towards achieving adequate state capacity with varying degrees of success. However, we argue that bringing governments “in the loop” is a critical step for effectively wielding this emerging resource on the world stage, navigating domestic developments wisely, and enabling meaningful leadership in the development of this historical technology.

Strengthen State Capacity

To enact this critical step, we suggest two policy objectives:

- **Measure capability consumption domestically.** In order to credibly and effectively wield this technology to further national interests, governments need to first and foremost become aware of the quantity of these newly manufactured resources. Governments may especially benefit from gaining insight into types of virtual labor that are of high geopolitical relevance, such as automated hacking or automated research, among others. Governments should have knowledge of, for instance, how much autonomous hacking is being minted domestically, and how much of this capability gets exported to foreign adversaries. They may also benefit from, for instance, being aware of the consumption of economically valuable capabilities, in order to gain a deeper understanding of the impacts on the labor market.
- **Pursue associated domestic and international opportunities.** With this enhanced awareness of the synthesis of these resources domestically, governments may benefit from building capacity for actively investigating the strategic opportunities enabled by this understanding. We hint at particular directions in the following section, yet here stress the policy objective of building the capacity to investigate such interventions in the first place. This may be easiest to achieve as part of existing organizations with expertise in related

domains, such as relevant public bodies (e.g., [US AISI](#)), and think tanks with a history of navigating the game theory of emerging technologies to advance national interests (e.g., [RAND](#)).

Leverage, Credibility, Security

As governments gain awareness of the quantity of these resources and actively pursue strategies informed by this understanding, we expect the following benefits to materialize:

- **Leverage.** This awareness may help highlight the scope and scale of this virtual workforce as a resource to be used in bargaining across the world stage. As nations boast extensive natural resources, advanced military equipment, or thriving consumer economies, so may the intellectual horsepower of these virtual entities constitute an additional negotiation token to be used in securing strategic deals. When composed with established policy practices, such as for instance through tariffs on imported quantities of virtual labor, these resources may provide a net increase in the range of available foreign policy actions.
- **Credibility.** Demonstrating awareness of domestic AI development through measures of capability consumption may help boost credibility both internationally, as well as among citizens impacted by the synthesis of this resource. Deploying mechanisms for proving the correctness and validity of these measurements can further provide the foundation of credibility needed to pursue landmark bilateral or multilateral initiatives on AI, analogous to how many nuclear security agreements have been predicated on literal [warhead counts](#) or [tons](#) of dual-use materials.
- **Security.** Similar to managing the proliferation of chemical precursors or radioactive materials for national security reasons, monitoring the quantities of dual-use AI capabilities being consumed by domestic organizations may help governments effectively address emerging threats. Beyond the domestic setting, surgical international initiatives may help extend existing non-proliferation efforts in the domain of dual-use AI capabilities, such as autonomous hacking.

Barriers Versus Setbacks

This extensive optionality may be achievable at little cost to companies developing AI. Consider that computational resources used in AI development are estimated to increase by [~5x per year](#), and the learning algorithms are estimated to become [~3x more efficient](#) in the same period, for a total increase of ~15x per year in the amount of virtual labor that can effectively be "minted." Assume that metering capability

consumption time incurs a 5% overhead relative to the baseline computational power of serving frontier models. This would be equivalent to a one-time setback of less than a week for the development of AI capabilities. That said, AI companies based in the US are estimated to be much farther ahead, potentially making it palatable to invest in the powerful geopolitical instruments detailed above, in order to be able to collect their future dividends.

To conclude, strengthening state capacity in AI by equipping governments with this enhanced awareness would allow them to wield these emerging resources more effectively in their dealings. By pursuing these policy objectives, governments can maintain leadership and ensure these powerful technologies serve national interests. We lay out a more concrete vision for how these remote measurement capabilities may be developed and applied in a dedicated resource on ["virtual diplomacy."](#)

[Noema Research](#) is an R&D lab working on techniques for remotely measuring AI capabilities with a view towards averting Great Power conflict. Based in the EU, we work with frontier labs and institutions from around the world to understand and manage the capability surface of AI systems. To enshrine this mission, leadership has formally pledged to donate 100% of proceeds on liquidity to charity.